

ALGORITMOS E LÓGICA DE PROGRAMAÇÃO

PROFESSORA DRA. LUCY MARI TABUTI

Exercícios de Operadores Aritméticos, Lógicos e Relacionais

1. Calcule o resultado das seguintes expressões aritméticas:

- a) $5 * 9 \bmod 2 - 4 / 2$
- b) $4 - 7 \operatorname{div} 2 + \operatorname{pot}(2,4) / \operatorname{rad}(4)$
- c) $\operatorname{rad}(\operatorname{pot}(3,3)) + \operatorname{pot}(2,3) \bmod \operatorname{rad}(4) - 8 / 2$
- d) $\operatorname{rad}(16) \operatorname{div} 9 / 3 - 4 * 3 - \operatorname{pot}(5 + 3, \operatorname{rad}(9))$
- e) $\operatorname{rad}(16 \bmod 4) + 5 * 2 / 10 - \operatorname{pot}(8 / 2, 2 \operatorname{div} 2)$
- f) $\operatorname{rad}(16) / 2 + 2 * 5 \bmod 3 / 1 * \operatorname{pot}(3, 2 + 4)$
- g) $\operatorname{rad}(\operatorname{rad}(16)) + \operatorname{pot}(\operatorname{pot}(2,2), \operatorname{pot}(2,3)) + 5 * 4 / 2$
- h) $\operatorname{rad}(\operatorname{rad}(\operatorname{pot}(2,4))) + \operatorname{pot}(\operatorname{rad}(4), \operatorname{pot}(\operatorname{rad}(4), 3)) + 9 - 5 * 4$
- i) $5 * 3 \bmod 5 + 4 \operatorname{div} 8 / 2 + 5 \bmod (4 + 2)$
- j) $\operatorname{rad}(100) \bmod \operatorname{pot}(4,3) - \operatorname{pot}(2,4) \operatorname{div} 8 / 2 + 5 - 3 * 2$

2. Calcule o resultado das seguintes expressões lógicas com operadores relacionais e aritméticos:

- a) $3 * 8 \bmod 2 = -20 / 4$
- b) $5 - 10 \operatorname{div} 3 > \operatorname{pot}(3,3) / \operatorname{rad}(25)$
- c) $\operatorname{rad}(\operatorname{pot}(3,2)) < \operatorname{pot}(5,2) \bmod \operatorname{rad}(49) - 81 / 9$
- d) $\operatorname{rad}(81) \operatorname{div} 8 / 3 - 4 \geq 5 - \operatorname{pot}(10 - 3, \operatorname{rad}(4))$
- e) $\operatorname{rad}(15 \bmod 3) + 15 * 12 / 6 \leq \operatorname{pot}(10 / 5, 12 \operatorname{div} 10)$
- f) $\operatorname{rad}(100) / 10 + 12 = 10 \bmod 25 / 5 * \operatorname{pot}(5, 2 + 1)$
- g) $\operatorname{rad}(\operatorname{rad}(81)) + \operatorname{pot}(\operatorname{pot}(2,3), \operatorname{pot}(2,2)) \leq 15 * 10 / 5$
- h) $\operatorname{rad}(\operatorname{rad}(\operatorname{pot}(3,4))) + \operatorname{pot}(\operatorname{rad}(16), \operatorname{pot}(\operatorname{rad}(4), 2)) \geq 19 - 15 * 2$

i) $27 * 25 \bmod 13 + 51 \operatorname{div} 3 > 12 + 35 \bmod (17 - 12)$

j) $\operatorname{rad}(36) \bmod \operatorname{pot}(2,3) - \operatorname{pot}(3,4) \operatorname{div} 18 / 6 < 25 - 13 * 2$

3. Calcule o resultado das seguintes expressões lógicas com operadores lógicos, relacionais e aritméticos:

a) $2 = 10 \bmod 2 \text{ e } -16 > 4$

b) $5 - 10 \operatorname{div} 3 > \operatorname{pot}(3,3) \text{ ou } \operatorname{rad}(25) = 5$

c) **não** F **e** $\operatorname{rad}(\operatorname{pot}(3,2)) < \operatorname{pot}(5,2) \text{ ou } 5 \bmod \operatorname{rad}(49) \geq 81 / 9$

d) **não** V **ou** $\operatorname{rad}(81) \operatorname{div} 8 / 3 - 4 \geq 5 - \operatorname{pot}(10 - 3, \operatorname{rad}(4))$

e) $\operatorname{rad}(15 \bmod 3) \leq 15 \text{ ou } 12 / 6 \leq \operatorname{pot}(10 / 5, 12 \operatorname{div} 10)$

f) $\operatorname{rad}(100) / 10 + 12 = 10 \bmod 25 \text{ ou } 5 * \operatorname{pot}(5,2 + 1) \text{ e não } V$

g) $\operatorname{rad}(\operatorname{rad}(81)) > 5 \text{ e } \operatorname{pot}(\operatorname{pot}(2,3), \operatorname{pot}(2,2)) \leq 15 \text{ e } 10 / 5 = 32$

h) $\operatorname{rad}(\operatorname{rad}(\operatorname{pot}(3,4))) > 3 \text{ ou } \operatorname{pot}(\operatorname{rad}(16), \operatorname{pot}(\operatorname{rad}(4),2)) \geq 19 \text{ ou } -15 * 2 = 7$

i) **não** F **ou** $27 * 25 \bmod 13 < 91 \text{ e } 51 \operatorname{div} 3 > 12 \text{ ou } 35 \bmod (17 - 12) \geq 13$

j) $\operatorname{rad}(36) \bmod \operatorname{pot}(2,3) \geq 10 \text{ ou não } V \text{ e } -\operatorname{pot}(3,4) \operatorname{div} 18 / 6 < 25 - 13 * 2$